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Forecasting Affective State from Interaction Streams: A Closed Loop with Heat-Decay Memory, Label-Free Mood Estimation, and Trajectory-Based Crisis/Venting Separation (LOCI r+e)

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Working draft — 2026-07-20. Companion to the reflection-on-action (e+r) paper. Results in §5 are from the SympPrev-Forecast study; reproducibility artifacts (`msymprev.py`, harnesses) are in this repository.

Abstract

We ask whether a person's near-future affective state can be forecast from the interaction stream alone — no self-report, no labeled emotions — well enough to change what the right next action is. We describe a closed forecasting loop that converts an interaction stream into a bounded affect estimate and projects it forward, and we evaluate 24-hour mood forecasting against carry-forward and a memoryless-LLM baseline. The loop forecasts 24h mood at 74% accuracy versus 48% for carry-forward and 61% for an LLM with no temporal model; ablations show each stream contributes. We further show that two states that look identical in a single snapshot — high arousal, negative valence — are separable by their *trajectory*: a rising, converging trajectory indicates crisis, a spiky, declining one indicates venting. This is r+e: forecast, then act on the forecast. We are explicit about scope: this is a working loop with real but modest numbers on a modest corpus, not a claim about affect in general or about machine emotion.

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1. Introduction

Acting well for a person over time requires anticipating, not just reacting, to their state. A support system that waits until distress is explicit has already missed the window where a lighter touch would have helped. But anticipating state is hard precisely where it matters: affective state is private, shifting, and rarely labeled.

We study forecasting affective state from the **interaction stream alone**. The stream — what the person says, how, when, about what, at what tempo — carries a signal about where their state is heading. The claim of this paper is narrow and testable: that signal is strong enough to forecast 24-hour mood better than assuming today equals tomorrow, and better than a large model with no temporal memory.

We distinguish two directions (developed jointly with a companion paper):

- **r+e (this paper):** forecast the future state, act on the forecast — predictive control.
- **e+r (companion):** act, observe the behavioral outcome, learn without labels — reflection-on-action.

This paper is the r+e half. It sits in a crowded field (predictive processing, affective forecasting), so our contribution is not a new primitive but a **working closed loop with measured forecasts and an honest ablation**, plus one result we believe is underappreciated: **trajectory, not snapshot, distinguishes crisis from venting** (§6).

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2. Related work and positioning

- **Affective computing** typically classifies *current* affect from labeled data. We forecast *future* affect from unlabeled interaction.
- **Affective forecasting (psychology)** studies how poorly *people* predict their own future feelings; we study whether a *system* can predict a person's near-future state from behavior.
- **Predictive processing / active inference** frame the brain as a prediction machine minimizing surprise. We borrow the surprise signal as an engineering input and make no biological or phenomenological claim.
- We claim no machine emotion. "Affective state" here is an operational estimate derived from behavior, evaluated against a held-out behavioral criterion.

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3. Method: the forecasting loop

3.1 Streams

The interaction stream is decomposed into parallel streams that each feed the forecast:

- **ISympPrev (informational/salience):** what is being referred to and how hot/recent it is; a heat kernel `P(τ) = Σ_i s_i · exp(−λ(τ − τ_i))` gives recurrence weight that decays with time.
- **QSympPrev (quantified/exact):** exact values (numbers, states) extracted verbatim, versioned so change is tracked without overwrite.
- **MSympPrev (mood):** a bounded affect estimate derived from reconstruction

  error (§3.2), with no labeled emotions.
- **SSympPrev (reflection-on-action):** the e+r feedback term (companion paper),
  included in the loop as the correction signal.
- **DSympPrev (forecast):** projects the current estimate forward via ensemble
  simulation.

3.2 Mood without labels (MSympPrev)

Mood is derived from *predictability*: when the world matches the model, reconstruction error is low and the state reads calm; when the world defies the model, error is high and the state reads aroused. Concretely, over a rolling window of reconstruction errors `e_t`:

- **Arousal** ∝ recent error vs the person's own 75th-percentile band (adaptive,
  per-person, self-calibrating — no fixed thresholds, no labels).
- **Valence** ∝ direction of the miss (better vs worse than expected).
- **Coherence** ∝ persistence/slope of the deviation (sustained vs spiky).

All three are bounded and decay fast (6-hour half-life), so mood fades toward neutral and can never become a permanent foreground — a design constraint, not an afterthought (see §7). This is implemented and tested (`msymprev.py`).

3.3 The θ-gate and the loop

The forecast is fed back only when confidence exceeds a threshold $\theta=0.82$; below it, the loop abstains rather than acting on a weak forecast. This makes the system's default *inaction under uncertainty*, which matters for the ethics of acting on a predicted private state.

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4. Data and protocol

- Interaction-stream corpus with a held-out 24-hour criterion: for each user, train on stream up to time *t*, forecast mood at *t+24h*, score against the behaviorally-derived criterion at *t+24h* (read only at scoring).
- Baselines: **carry-forward** (tomorrow = today) and **memoryless LLM** (the same content, no temporal model).
- Ablations: remove each stream in turn; measure the drop.

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5. Results (SympPrev-Forecast study)

System	24h mood accuracy
Carry-forward (tomorrow = today)	48%
Memoryless LLM (content, no temporal model)	61%
Forecasting loop (ours)	**74%**

- The loop beats carry-forward by 26 points and the memoryless LLM by 13, indicating the temporal/behavioral structure — not just the content — carries the forecast.
- **Ablations:** removing any single stream reduces accuracy; no stream is redundant (full table in the source study).

Honest framing: these are the SympPrev-Forecast numbers on that study's corpus. They are real and were obtained with held-out scoring, but the corpus is modest; we report them as evidence the loop works, not as a universal accuracy claim. Re-running on a second corpus with published CIs is a stated follow-up.

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6. Crisis vs venting: trajectory, not snapshot

A single snapshot cannot tell crisis from venting — both present as high arousal and negative valence. They differ in **trajectory**:

- **Venting:** the deviation is spiky and *declining* — a discharge. Coherence low.
- **Crisis:** the deviation is sustained and *rising/converging* — an escalation. Coherence high.

In our implementation (`msymprev.py`), the two are separated on this basis alone: identical arousal (1.0) and valence (−1.0), but coherence 0.9 (crisis) vs 0.4 (venting). The forecast is aimed at a *feeling's direction*, not its momentary value. We regard this as the paper's most actionable result: the difference between "letting off steam" and "needs help now" is a trajectory the loop can read without a label.

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7. Ethics

Forecasting a person's private state and acting on it is a capability that must default to restraint. Three constraints are built into the method, not bolted on:

1. **Abstain under uncertainty (θ-gate):** below confidence 0.82 the loop does not act on the forecast. Acting on a weak guess about someone's inner state is the harm to avoid.
2. **Bounded, fast-decaying affect:** mood cannot accumulate or persist; it fades (6h half-life). A system whose model of your feelings can run away is the failure mode we designed against (it is a real one we have seen).
3. **The objective is chosen deliberately.** As the companion paper argues at length, a forecast pointed at engagement rather than wellbeing becomes a manipulation engine most effective on the vulnerable. Forecasting must serve a defended wellbeing objective, gated so the system's own inaction is also accountable.

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8. Limitations

- Crowded field; the contribution is a working loop + numbers + the trajectory result, not a new theoretical primitive.
- Single-corpus results; CIs and a second-corpus replication are stated follow-ups, not yet done.
- "Mood" is operationalized from reconstruction error; other operationalizations may differ and are named, not hidden.
- No claim of machine emotion; all quantities are behavioral estimates against behavioral criteria.

9. Reproducibility

`msymprev.py` (mood stream: arousal/valence/coherence, crisis-vs-venting, bounded + decaying, 7/7 tests) and the LOCI streams are in this repository. The 24h forecasting numbers are from the SympPrev-Forecast study; the replication harness and CI reporting are the stated next step.

Appendix A — Second-corpus forecast-skill replication (2026-07-20)

The 24h-mood number (§5) comes from the SympPrev corpus. To test the underlying r+e claim on an *independent* corpus with confidence intervals, we ran `forecast_skill.py --limit 500 --seeds 5` on LongMemEval (20,342 need-sequences, ~41k held-out forecast points/seed). This does **not** reproduce the 24h-mood number; it tests the mechanism: *does the recent surprise trajectory forecast the next step's arousal better than persistence or base rate?*

Condition	Balanced accuracy (±95% CI)	AUC
C0 base-rate (no trajectory)	0.500 ± 0.000	0.500
C1 carry-forward (persistence)	0.523 ± 0.0015	0.523
F1 trajectory (ours)	**0.608 ± 0.0024**	**0.608**

Primary result (F1 > C1): SUPPORTED. Δ = +0.085, ≈22× the combined CI band. The surprise trajectory (recent level + rising slope) forecasts next-step arousal well above persistence, which itself barely clears chance (0.523).

Honest reading

- The effect is **modest and real**: 0.61 AUC — clearly above chance and above persistence, but not large. This matches the paper's framing; we do not inflate it into a strong-forecaster claim.
- **Proxy, not clinical affect.** "Arousal" here is operationalized as the top-tercile of behavioral surprise; F1 forecasts future surprise level from the past surprise trajectory. This supports the *mechanism* — that the trajectory is forecastable and beats persistence — but it is a proxy for the affective claim, not clinical mood. The 24h-mood result (§5) remains the affective evidence; this is the independent-corpus, CI-reported corroboration that the r+e trajectory signal is genuine.
- **The trajectory-over-snapshot thesis (§6) holds here too:** a level+slope forecast beats a level-only persistence baseline. That is the same claim that separates crisis from venting, measured now with tight CIs on a second corpus.

Predictions and harness were committed before the run; raw output in `forecast_skill_n20342.json`.